



KWAZULU-NATAL PROVINCE

EDUCATION
REPUBLIC OF SOUTH AFRICA

**NATIONAL
SENIOR CERTIFICATE**

GRADE 10

**PHYSICAL SCIENCES P2
FINAL EXAMINATION
NOVEMBER 2025**

Stanmorephysics.com

MARKS: 100

DURATION: 2 hours

This question paper consists of 10 pages and 2 data sheets.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SEVEN questions. Answer ALL the questions in the ANSWER BOOK.
2. Start EACH question on a NEW page in the ANSWER BOOK.
3. Number the answers correctly according to the numbering system used in this question paper.
4. Leave ONE line between two sub-questions, for example between QUESTION 2.1 and QUESTION 2.2.
5. You may use a non-programmable calculator.
6. You may use appropriate mathematical instruments.
7. You are advised to use the attached DATA SHEETS.
8. Show ALL formulae and substitutions in ALL calculations.
9. Round off your final numerical answers to a minimum of TWO decimal places.
10. Give brief motivations, discussions et cetera where required.
11. Write neatly and legibly.

QUESTION 1: MULTIPLE-CHOICE QUESTIONS

Four options are provided as possible answers to the following questions. Each question has only ONE correct answer. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.5) in the ANSWER BOOK, for example 1.11 E.

1.1 Which one of the following is an example of a pure substance?

- A Carbon dioxide
- B Air
- C Steel
- D Aqueous sodium chloride

(2)

1.2 Which one of the following has the highest first ionization energy?

- A Li
- B F
- C Na
- D Cl

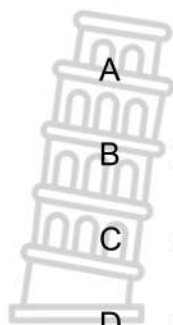
(2)

1.3 Which of the following is true for the process of condensation?

- A The substance is heated and changes phase to gas.
- B The substance is heated and changes phase to liquid.
- C The substance is cooled and changes phase to gas.
- D The substance is cooled and changes phase to liquid.

(2)

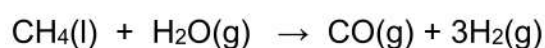
1.4 Which one of the following contains $6,02 \times 10^{23}$ ATOMS?



- A 10 g of Ne
- B 28 g of N_2 gas
- C $4,48 \text{ dm}^3$ of CH_4 gas at STP
- D $22,4 \text{ dm}^3$ of NH_3 gas at STP

(2)

1.5 Consider the balanced chemical equation for the reaction below:



The number of CH_4 molecules required to form $22,4 \text{ dm}^3$ of H_2 at STP is ...

- A $6,02 \times 10^{23}$
- B $\frac{1}{2} \times 6,02 \times 10^{23}$
- C $\frac{1}{3} \times 6,02 \times 10^{23}$
- D $\frac{1}{4} \times 6,02 \times 10^{23}$

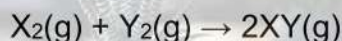


(2)

[2 x 5 = 10]

QUESTION 2 (Start on a new page.)

- 2.1 Two hypothetical gases, X_2 and Y_2 , react in a closed container to form XY gas according to the equation:



After the reaction, the container is found to contain both XY gas and some unreacted Y_2 gas.

- 2.1.1 Would you classify the contents of the container, after the reaction, as a MIXTURE or a PURE SUBSTANCE? (1)

- 2.1.2 To separate XY from Y_2 the following steps take place:

STEP 1: The contents of the container is cooled to liquify BOTH gases.

STEP 2: The liquid is thereafter gradually heated to a maximum temperature of $-60\text{ }^{\circ}\text{C}$, to separate Y_2 and XY

The boiling points of Y_2 and XY is shown in the table below:

Substance	Boiling Point ($^{\circ}\text{C}$)
Y_2	-180
XY	-50

- (a) At which temperature will both gases be in their liquid phase.
Choose from: $20\text{ }^{\circ}\text{C}$, $-70\text{ }^{\circ}\text{C}$, $-190\text{ }^{\circ}\text{C}$. (1)
- (b) Which of the two substances, Y_2 or XY , will more easily be converted to gas when the temperature is increased in STEP 2. (1)
- (c) Does STEP 2 represent a PHYSICAL or CHEMICAL change?
Explain the answer. (2)
- (d) The substance that is more easily converted to gas is collected. It is observed that the collected gas is not 100% pure.

Give a reason for the observation. (2)

2.2 Ammonium sulphate is one of the most common fertilizers used worldwide.

Ammonia reacts with sulphuric acid (hydrogen sulphate) to form ammonium sulphate.

2.2.1 Write a balanced chemical equation for the above reaction. (3)

2.2.2 Is the above reaction an example of a SYNTHESIS or DECOMPOSITION reaction? (1)

2.3 A farmer chooses to use ammonium nitrate (NH_4NO_3) as a fertilizer. He needs to supply 30 kg of NITROGEN to a field.

2.3.1 Calculate the percentage nitrogen (N) in ammonium nitrate (NH_4NO_3). (3)

2.3.2 Determine the mass of ammonium nitrate (NH_4NO_3) required. (3)

2.4 Use the balanced chemical equation below to prove the law of conservation of mass.



(3)

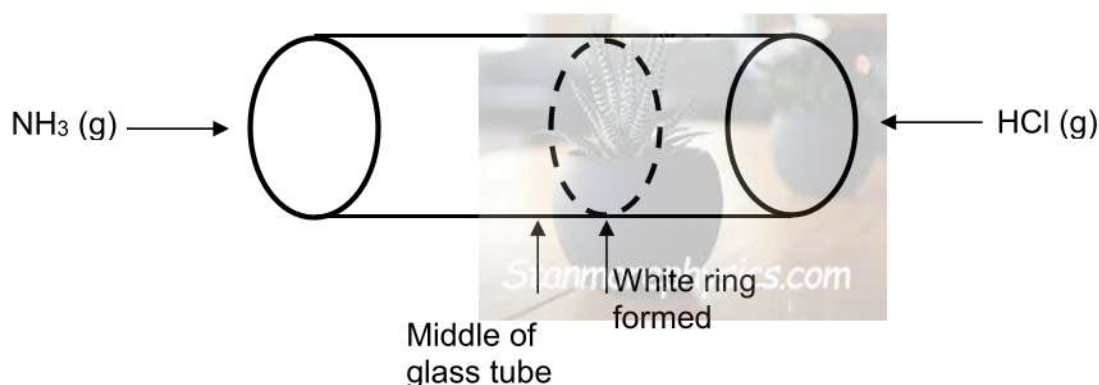
[20]

QUESTION 3 (Start on a new page.)

3.1 A teacher performs a demonstration using ammonia (NH_3) and hydrochloric acid (HCl) GASES in a long glass tube.

Ammonia and hydrochloric acid are simultaneously released from opposite ends of a long glass tube. A white ring of ammonium chloride (NH_4Cl) form where the gases react.

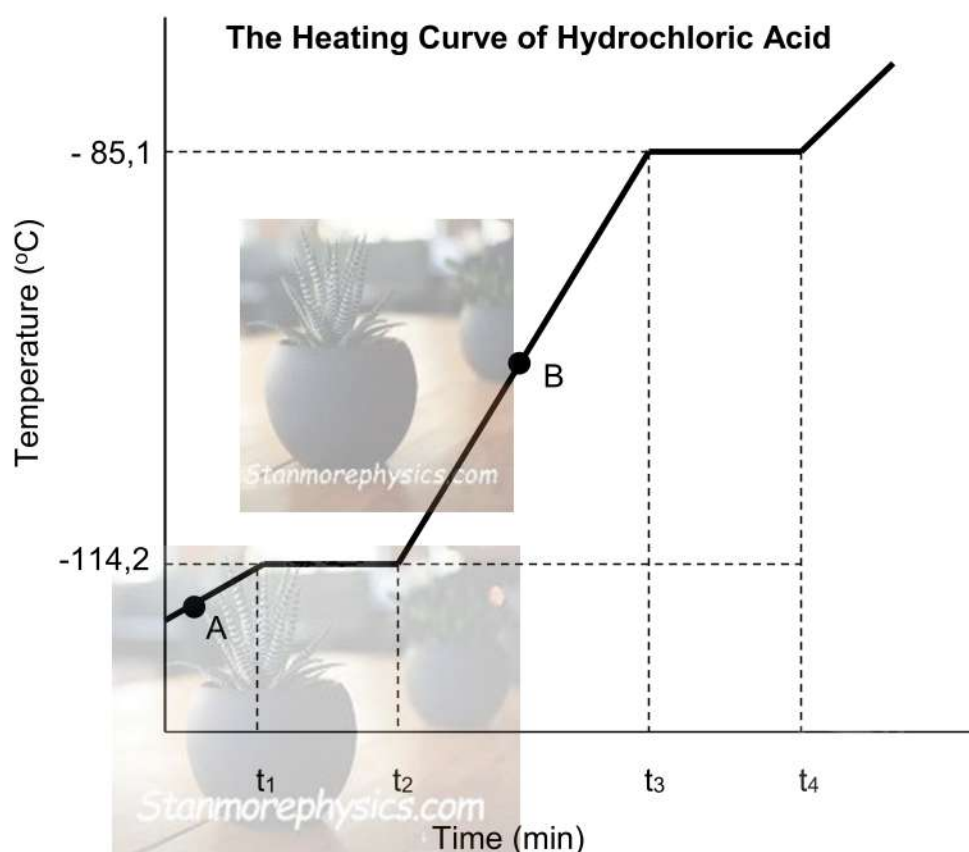
The position of the white ring formed is shown on the diagram below.





- 3.1.1 Define the term *diffusion* in words. (2)
- 3.1.2 Provide a reason why the white ring formed closer to HCl than NH₃. (1)
- 3.1.3 The temperature of the gases in the glass tube is now increased.
How would this effect the TIME taken for the white ring to appear?
Choose from INCREASE, DECREASE or REMAIN THE SAME.
Explain the answer using the kinetic molecular theory. (3)

- 3.2 The heating curve of Hydrochloric acid (HCl) is represented by the graph below. The graph is not drawn to scale.



- 3.2.1 Define the term *boiling point* in words. (2)
- 3.2.2 Write down the boiling point of hydrochloric acid. (1)
- 3.2.3 At which point, A or B, are the forces between HCl particles stronger? (1)
- 3.2.4 Refer to the temperature, energy and movement of particles to explain the changes that take place between t₂ and t₃. (3)

[13]

QUESTION 4 (Start on a new page.)

4.1 Silicon occurs naturally as three stable isotopes:

- Si – 28
- Si – 29
- Si – 30

The percentage abundance of Si - 28 is 92,2 % and the relative atomic mass of silicon is 28,1 AMU.

4.1.1 Define the term isotope in words. (2)

4.1.2 Determine the percentage abundance of Si – 29. (5)

4.1.3 Fill in the table below for the most abundant isotope of silicon. Write down only the letter (a – c) and the correct answer.

Number of protons	(a)
Number of neutrons	(b)
Number of valence electrons	(c)

(3)

4.2 Magnesium combines with chlorine to form magnesium chloride.

4.2.1 Identify the type of bond that forms when magnesium and chlorine combine. (1)

4.2.2 Draw the Lewis dot diagram for magnesium chloride. (2)

4.2.3 Draw the Aufbau diagram for the magnesium ion. (2)

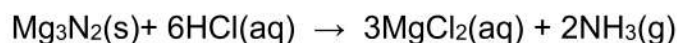
4.2.4 Write down the name of the noble gas that has the same electronic configuration as the chloride ion. (1)

4.3 How does the atomic radius of Chlorine (Cl) compare to Magnesium (Mg)? Choose from GREATER THAN, LESS THAN or EQUAL TO. Explain your answer by referring to the atomic structure. (4)

[20]

QUESTION 5 (Start on a new page)

Magnesium nitride (Mg_3N_2) reacts with hydrochloric acid (HCl) according to the balanced equation below:



In one such reaction a learner completely reacts 10 g magnesium nitride (Mg_3N_2) with $1,2 \text{ dm}^3$ hydrochloric acid (HCl) at standard temperature and pressure (STP).

5.1 Define the term *concentration* in words. (2)

5.2 Calculate the number of moles of Mg_3N_2 that reacted. (3)

5.3 Determine the:

5.3.1 Concentration of HCl (4)

5.3.2 Volume of NH_3 gas (in dm^3) that formed at STP (4)

5.3.3 Number of chloride ions (Cl^-) present in the MgCl_2 produced (5)

[18]

QUESTION 6 (Start on a new page)

6.1 Analysis of a compound that is made up of Magnesium (Mg), Phosphorus (P) and Oxygen (O) only, provided the following percentage composition by mass.

- Magnesium: 27,7 %
 - Phosphorus: 23,6 %
 - Oxygen: 48,7 %
- (6)

Determine the empirical formula of this compound.

6.2 Determine the mass of hydrated potassium sulphate ($\text{K}_2\text{SO}_4 \cdot 7\text{H}_2\text{O}$) needed to prepare 100 ml of a $0,2 \text{ mol} \cdot \text{dm}^{-3}$ potassium sulphate solution. (4)

[10]

QUESTION 7 (Start on a new page)

A metal **Y** reacts with copper (ii) oxide (CuO) according to balanced chemical below:



A 5 g sample of **Y** reacts with excess copper (ii) oxide (CuO) to produce 15 g of copper (Cu).

The percentage yield of copper (Cu) in this reaction is 85 %.

- 7.1 Determine the theoretical mass of copper (Cu) formed. (3)
- 7.2 Use relevant calculations to determine the NAME of metal **Y**. (6)

[9]



DATA FOR PHYSICAL SCIENCES GRADE 10

CHEMISTRY

TABLE 1: PHYSICAL CONSTANTS

NAME	SYMBOL	VALUE
Standard pressure	p^θ	$1,013 \times 10^5 \text{ Pa}$
Molar gas volume at STP	V_m	$22,4 \text{ dm}^3 \cdot \text{mol}^{-1}$
Standard temperature	T^θ	273 K
Avogadro's constant	N_A	$6,02 \times 10^{23} \text{ mol}^{-1}$

TABLE 2: FORMULAE

$n = \frac{m}{M}$	$n = \frac{N}{N_A}$
$c = \frac{n}{V}$ or $c = \frac{m}{MV}$	$n = \frac{V}{V_m}$



TABLE 3: THE PERIODIC TABLE OF ELEMENTS

1 (I)	2 (II)	3	4	5	6	7	8	9	10	11	12	13 (III)	14 (IV)	15 (V)	16 (VI)	17 (VII)	18 (VIII)
1 2,1 H 1																	2 He 4
3 1,0 Li 7	4 1,5 Be 9											5 2,0 B 11	6 2,5 C 12	7 3,0 N 14	8 3,5 O 16	9 4,0 F 19	10 Ne 20
11 0,9 Na 23	12 1,2 Mg 24											13 1,5 Al 27	14 1,8 Si 28	15 2,1 P 31	16 2,5 S 32	17 3,0 Cl 35,5	18 Ar 40
19 0,8 K 39	20 1,0 Ca 40	21 1,3 Sc 45	22 1,5 Ti 48	23 1,6 V 51	24 1,6 Cr 52	25 1,5 Mn 55	26 1,8 Fe 56	27 1,8 Co 59	28 1,8 Ni 59	29 1,9 Cu 63,5	30 1,6 Zn 65	31 1,6 Ga 70	32 1,8 Ge 73	33 2,0 As 75	34 2,4 Se 79	35 2,8 Br 80	36 Kr 84
37 0,8 Rb 86	38 1,0 Sr 88	39 1,2 Y 89	40 1,4 Zr 91	41 Nb 92	42 1,8 Mo 96	43 1,9 Tc	44 2,2 Ru 101	45 2,2 Rh 103	46 2,2 Pd 106	47 1,9 Ag 108	48 1,7 Cd 112	49 1,7 In 115	50 1,8 Sn 119	51 1,9 Sb 122	52 2,1 Te 128	53 2,5 I 127	54 Xe 131
55 0,7 Cs 133	56 0,9 Ba 137	57 La 139	72 1,6 Hf 179	73 Ta 181	74 W 184	75 Re 186	76 Os 190	77 Ir 192	78 Pt 195	79 Au 197	80 Hg 201	81 1,8 Tl 204	82 1,8 Pb 207	83 1,9 Bi 209	84 2,0 Po	85 2,5 At	86 Rn
87 0,7 Fr	88 0,9 Ra 226	89 Ac															
58 Ce 140	59 Pr 141	60 Nd 144	61 Pm	62 Sm 150	63 Eu 152	64 Gd 157	65 Tb 159	66 Dy 163	67 Ho 165	68 Er 167	69 Tm 169	70 Yb 173	71 Lu 175				
90 Th 232	91 Pa	92 U 238	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr				

KEY/SLEUTEL

Electronegativity
ElektronegatiwiteitAtomic number
AtoomgetalSymbol
SimboolApproximate relative atomic mass
Benaderde relatiewe atoommassa



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GRADE 10

**PHYSICAL SCIENCES P2
FINAL EXAMINATION
MARKING MEMORANDUM**

NOVEMBER 2025 [orephysics.com](https://www.stanmorephysics.com)

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MARKS: 100

DURATION: 2 hours

QUESTION 1

- | | | |
|-----|------|-------------|
| 1.1 | A ✓✓ | (2) |
| 1.2 | B ✓✓ | (2) |
| 1.3 | D ✓✓ | (2) |
| 1.4 | C ✓✓ | (2) |
| 1.5 | C ✓✓ | (2) |
| | | [10] |

QUESTION 2

- 2.1.1 Mixture ✓ (1)
- 2.1.2a) - 190°C ✓ (1)
- 2.1.2b) Y₂ ✓ (1)
- 2.1.2c) Physical (change). ✓
No new substance is formed. OR No chemical reaction took place. ✓ (2)
- 2.1.2d) Some of the XY would evaporate below its boiling point. ✓ ✓ (2)
- 2.2.1 $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$  LHS ✓
RHS ✓
Balancing ✓ (3)
- 2.2.2 Synthesis ✓ (1)
- 2.3.1 $\% \text{ N} = \frac{14 \times 2}{14 \times 2 + 1 \times 4 + 16 \times 3} \times 100$ ✓
= 35 % ✓ (3)
- 2.3.2 **POSITIVE MARKING FROM 2.3.1**
 $m(\text{NH}_4\text{NO}_3) = 30 \div 0,35$ ✓
= 85,71 kg ✓ (3)
- 2.4 Mass of reactants = $4[14 + 1 \times 3] + 5[16 \times 2]$ ✓ = 228 g
Mass of products = $4[14 + 16] + 6[1 \times 2 + 16]$ ✓ = 228 g
Mass of reactants is equal to the mass of products. Therefore mass is conserved. ✓ (3)

[20]

QUESTION 3

- 3.1.1 The movement of atoms or molecules ✓ from an area of higher concentration to an area of lower concentration. ✓ (2)
- 3.1.2 Ammonia diffuses faster than hydrochloric acid ✓ (1)
- 3.1.3 Decrease. ✓
At a higher temperature the particles move faster OR have more kinetic energy ✓.
Therefore, the gases diffuse faster. ✓ (3)
- 3.2.1 The temperature of a liquid at which its vapour pressure equals the external (atmospheric) pressure. ✓✓ (2)
- 3.2.2 - 85,1 °C ✓ (1)
- 3.2.3 (point) A ✓ (1)
- 3.2.4 The temperature of HCl increase. ✓ The average kinetic energy of the particles increases. ✓ Particles move faster. ✓ (3)

[13]

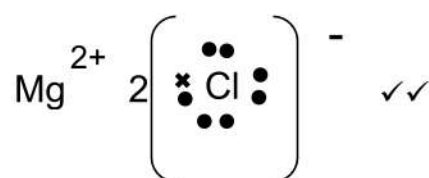
QUESTION 4

- 4.1.1 Atoms of the same element having the same number of protons, but different number of neutrons. ✓✓ (2)
- 4.1.2 Let the % of Si – 29 be X.
The % of Si – 30 will be 7,8 – X
$$28,1 \checkmark = 28 \times \frac{92,2}{100} \checkmark + 29 \times \frac{X}{100} \checkmark + 30 \times \frac{7,8 - X}{100} \checkmark$$
$$X = 5,6 \% \checkmark$$
 (5)
- 4.1.3 (a) = 14 ✓
(b) = 14 ✓
(c) = 4 ✓ (3)
- 4.2.1 Ionic bonds ✓ (1)
- 4.2.2 (2)

OPTION 1

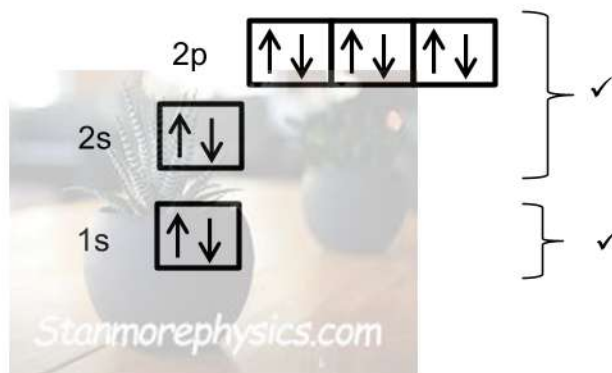


OPTION 2



4.2.3

(2)



4.2.4 Argon ✓

(1)

4.3 Less than. ✓

- Both chlorine and magnesium are in the same period OR have the same number of energy levels. ✓
- The effective nuclear charge of chlorine is higher than magnesium. ✓
- The force of attraction between the nucleus and the outer orbital is stronger in chlorine. ✓

(4)

[20]

QUESTION 5

5.1 The number of moles of solute per cubic decimetre of solution. ✓✓

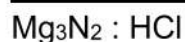
(2)

5.2

$$\begin{aligned} n(\text{Mg}_3\text{N}_2) &= \frac{m}{M} \checkmark \\ &= \frac{10}{24 \times 3 + 14 \times 2} \checkmark \\ &= 0,1 \text{ mol } \checkmark \end{aligned}$$

(3)

5.3.1 **POSITIVE MARKING FROM 5.2**



$1 : 6 \checkmark$

$n(\text{HCl}) = 0,6 \text{ mol}$

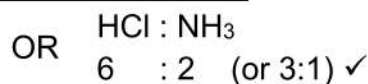
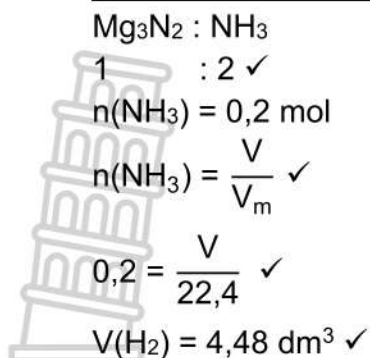
$c(\text{HCl}) = \frac{n}{V} \checkmark$

$= \frac{0,6}{1,2} \checkmark$

$= 0,5 \text{ mol} \cdot \text{dm}^3 \checkmark$

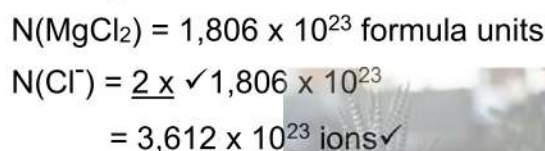
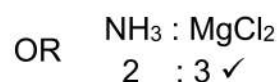
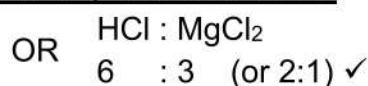
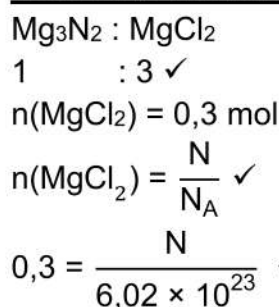
(4)

5.3.2 **POSITIVE MARKING FROM 5.2 AND 5.3.1**



(4)

5.3.3 **POSITIVE MARKING FROM 5.2, 5.3.1 AND 5.3.2**



(5)

[18]

QUESTION 6

6.1

Marking criteria:

- Calculate the number of moles of each element ($3 \times 1 = 3$) ✓✓✓
- Determine the simplest ratio of all elements ✓
- Multiplying by 2 ✓
- Final answer: $\text{Mg}_3\text{P}_2\text{O}_8$ ✓

Element	mass (g)	$n = \frac{m}{M}$	Simplest Ratio			
Mg	27,7	$\frac{27,7}{24} = 1,15 \checkmark$	$\frac{1,15}{0,76} = 1,51$	✓	x 2 ✓	3
P	23,6	$\frac{23,6}{31} = 0,76 \checkmark$	$\frac{0,76}{0,76} = 1$			2
O	48,7	$\frac{48,7}{16} = 3,04 \checkmark$	$\frac{3,04}{0,76} = 4$			8

Empirical Formula: $\text{Mg}_3\text{P}_2\text{O}_8$ ✓

(6)

6.2.1 $M(K_2SO_4 \cdot 7H_2O) = 39 \times 2 + 32 + 16 \times 4 + 7 \times 18$
 $= 300 \text{ g.mol}^{-1}$

OPTION 1



$$c = \frac{m}{MV} \checkmark$$

$$0,2 = \frac{m}{300(0,1)} \checkmark$$

$$m = 6 \text{ g} \checkmark$$

OPTION 2

$$c = \frac{n}{V} \checkmark$$

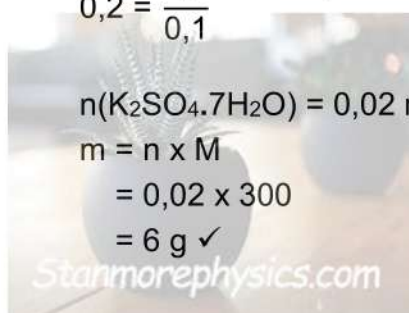
$$0,2 = \frac{n}{0,1}$$

$$n(K_2SO_4 \cdot 7H_2O) = 0,02 \text{ mol}$$

$$m = n \times M$$

$$= 0,02 \times 300$$

$$= 6 \text{ g} \checkmark$$



(4)
[10]

QUESTION 7

7.1 $\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$

$$85 \checkmark = \frac{15 \checkmark}{\text{theoretical yield}} \times 100$$

$$m(\text{Cu})_{\text{theoretical}} = 17,65 \text{ g} \checkmark$$

OR

$$m(\text{Cu})_{\text{actual}} = m(\text{Cu})_{\text{theoretical}} \times 0,85$$

$$15 \checkmark = m(\text{Cu})_{\text{theoretical}} \times 0,85 \checkmark$$

(3)

7.2 **POSITIVE MARKING FROM 7.1**

Marking criteria:

- Divide by 63,5 $\checkmark$
- Ratio (2:3) $\checkmark$
- Formula: $n(Y) = \frac{m}{M} \checkmark$
- Substitute $n(Y)$ $\checkmark$
- Substitute 5g $\checkmark$

Final answer: Aluminium $\checkmark$

$$n(\text{Cu}) = \frac{m}{M}$$

$$= \frac{17,65}{63,5} \checkmark$$

$$= 0,278 \text{ mol}$$

Y : Cu

2 : 3 $\checkmark$

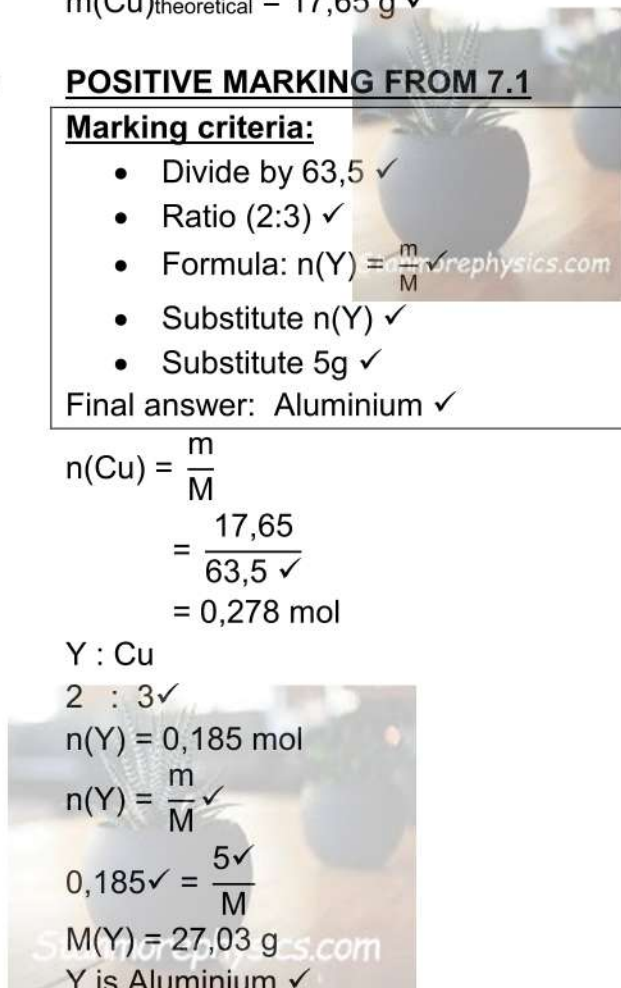
$$n(Y) = 0,185 \text{ mol}$$

$$n(Y) = \frac{m}{M} \checkmark$$

$$0,185 \checkmark = \frac{5 \checkmark}{M}$$

$$M(Y) = 27,03 \text{ g}$$

Y is Aluminium $\checkmark$



(6)

[9]

100

TOTAL: